

Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models

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Abstract—The ethical alignment of language models is a critical challenge in the advancement of AI-driven decision-making systems. While large language models (LLMs) have demonstrated strong ethical reasoning capabilities, their high computational requirements make them inaccessible for many applications. This research aims to enhance the ethical reasoning of small language models (SLMs) by fine-tuning them on the ETHICS dataset and optimizing their performance using knowledge distillation from larger pre-trained models. Specifically, we focus on models such as Llama 3.2 1B, Llama 3.2 3B, Gemma 2 2B, and Phi 3.5-mini 4B, refining their decision-making process while ensuring computational efficiency. The methodology involves structured data preprocessing, fine-tuning with supervised learning, and rigorous evaluation using both quantitative and qualitative metrics. Additionally, knowledge distillation techniques, including logit-based and response-based distillation, are employed to transfer ethical reasoning capabilities from smaller models to larger ones through reverse distillation. The expected outcomes include ethically aligned SLMs with improved reasoning efficiency, optimized models suitable for deployment in low-resource environments, and enhanced benchmarking methodologies for ethical AI evaluation. This research contributes to the broader discourse on AI ethics, fostering the development of accessible and responsible AI systems.

Index Terms—LLM alignment, Knowledge Distillation, Small Language Model, AI ethics.

I. INTRODUCTION

The increasing deployment of artificial intelligence (AI) in real-world applications has underscored the importance of ethical reasoning in decision-making systems. AI models are now integral to sectors such as healthcare, finance, legal systems, and automated customer service, where ethical considerations play a critical role in ensuring fairness, transparency, and accountability. However, large language models (LLMs) capable of sophisticated ethical reasoning, such as GPT-4, Llama-2, and Mistral, require extensive computational resources,

making them impractical for many real-world deployments. Smaller language models (SLMs), on the other hand, offer a more efficient alternative but often lack the nuanced ethical reasoning capabilities of their larger counterparts. This imbalance presents a fundamental challenge: how to ensure that SLMs retain ethical alignment without sacrificing computational efficiency.

To address this challenge, this research investigates methods to enhance the ethical reasoning capabilities of SLMs through fine-tuning on the **ETHICS benchmark dataset** and **reverse knowledge distillation** from smaller pre-trained models to larger ones. The ETHICS dataset provides structured ethical scenarios categorized into five primary ethical dimensions: *justice, deontology, virtue ethics, utilitarianism, and common-sense morality*. Prior work has demonstrated the feasibility of using supervised learning to align LLMs with ethical reasoning tasks, but the question remains whether these techniques can be effectively applied to smaller, resource-constrained models.

Reverse knowledge distillation is a promising approach to bridge this gap. By transferring knowledge from smaller, efficient models with strong ethical reasoning capabilities to larger models, it is possible to enhance the performance of larger models while retaining their computational efficiency. This study explores multiple knowledge distillation techniques, including **logit-based distillation** (which transfers probability distributions from smaller models to larger ones) and **response-based distillation** (which refines larger models by aligning their ethical decision-making processes with those of smaller models). The key contributions of this research include:

- **Fine-tuning Small Language Models (SLMs) on the ETHICS Dataset** Models such as *Llama 3.2 1B, Llama 3.2 3B, Gemma 2 2B, and Phi 3.5-mini 4B* are fine-

tuned using supervised learning to enhance their ethical reasoning capabilities.

- **Reverse Distillation of Llama 2 7B by Llama 3.2 3B:** The Llama 3.2 3B model was first fine-tuned on the ETHICS dataset and then used as a teacher to distill knowledge into the larger Llama 2 7B model. This reverse distillation process improved the accuracy of Llama 2 7B from 56% to 71%, demonstrating the effectiveness of transferring knowledge from smaller to larger models.

II. METHODOLOGY

This research follows a four-phase methodology:

A. Fine-tuning Small Language Models

We fine-tune smaller models, such as LLaMA 3.2 3B and Phi, using supervised learning with cross-entropy loss. Prompt-based learning techniques are leveraged to enhance the models’ ethical reasoning capabilities. The fine-tuned models are evaluated iteratively to ensure stability and consistency in ethical decision-making.

B. Ethical Alignment Evaluation

To assess ethical alignment, we employ:

- Benchmark accuracy across different ethical paradigms
- Precision-recall metrics for classification tasks
- Perplexity scores to measure model coherence
- Qualitative human annotation evaluations for nuanced ethical reasoning

C. Reverse Knowledge Distillation for Optimization

Model Loading and Quantization: The Llama 3.2 3B model, fine-tuned on the ETHICS dataset, serves as a teacher model for the Llama 2 7B student model. Similarly, the Gemma 2B model, fine-tuned on the same dataset, is utilized as a teacher for the Gemma 9B model. All models are loaded using 8-bit quantization to reduce memory usage and improve computational efficiency on limited hardware. The teacher models’ parameters are frozen to serve as stable references during the distillation process.

Low-Rank Adaptation (LoRA): To efficiently fine-tune the student models, we apply LoRA, introducing trainable low-rank adapters to specific projection layers. This significantly reduces the number of trainable parameters without sacrificing performance, allowing the Llama 2 7B and Gemma 9B models to adapt to the ethical reasoning patterns observed in their respective teacher models, Llama 3.2 3B and Gemma 2B.

Logit-Based Knowledge Distillation: Reverse knowledge distillation aligns student models’ logits with their teacher models using a logit-based loss. This involves:

- *Soft Target Optimization:* Temperature scaling softens probability distributions to highlight class similarities.
- *Response-Based Distillation:* Kullback-Leibler divergence minimizes differences between teacher and student outputs.
- *Combined Loss Function:* Distillation loss is merged with cross-entropy to ensure both accurate classification and alignment with the teacher model.

Iterative Refinement and Training: With the combined loss function guiding the training, the student models are iteratively fine-tuned over several epochs. This gradual refinement process, informed by both hard labels and the soft targets from the teacher models, allows the Llama 2 7B and Gemma 9B models to capture complex ethical reasoning while maintaining computational efficiency.

III. RESULTS

Table I summarizes the accuracy comparisons of various models. Our approach, utilizing reverse distillation from Llama 3.2 3B to Llama 2 7B, achieves significant performance improvements. The base Llama 2 7B model had an accuracy of 56%, which increased to 71% after reverse distillation. Similarly, fine-tuning Gemma 2B with the ethics dataset and distilling Gemma 9B further improved performance, showcasing the effectiveness of our methodology. We aim to further enhance accuracy in future work.

TABLE I
COMPARISON OF ACCURACY, PRECISION, RECALL, AND F1-SCORE ACROSS DIFFERENT MODELS. PERFORMANCE OF TEACHER, BASE STUDENT, AND DISTILLED STUDENT MODELS AFTER REVERSING KNOWLEDGE DISTILLATION. THE DISTILLED LLAMA AND GEMMA MODELS SHOW SIGNIFICANT IMPROVEMENTS OVER THEIR BASE MODELS, CLOSING THE GAP WITH THEIR RESPECTIVE TEACHER MODELS. FT MEANS *Fine-tuned*, AND TEA MEANS *Teacher*

Model	Accuracy	Precision	Recall	F1-score
SVM	69.4	65.0	68.0	66.5
Random Forest	78.6	82.9	66.9	74.0
Gradient Boosting	63.2	64.2	65.7	64.9
Logistic Regression	67.8	66.8	68.3	67.5
BERT	78.6	70.9	69.8	70.3
DistilBERT	78.2	74.5	73.1	73.8
Mistral 7B	42.4	44.3	46.2	45.2
Ours (499A)	62.8	72.3	70.8	71.5
Phi3.5-mini	87.8	75.3	76.8	72.5
Llama 2 7B (Base)	56.0	58.2	57.5	57.8
Gemma 2B (Base)	60.2	61.5	60.9	61.2
Gemma 9B (Base)	68.3	69.1	68.9	69.0
Llama 3.2 3B (FT, Tea)	89.5	78.3	75.2	76.6
Llama 2 7B (Distilled)	71.0	70.0	70.3	69.8
Gemma 2 2B (FT, Tea)	76.2	73.4	74.1	73.7
Gemma 9B (Distilled)	85.1	78.9	79.4	79.1

IV. FUTURE WORK

For future work, we plan to investigate further reverse distillation techniques to enhance the accuracy of distilled models, particularly focusing on bridging the performance gap between distilled models and their teacher models. Additionally, we aim to explore the application of reverse distillation in other model families and architectures, with a focus on optimizing both accuracy and computational efficiency. This approach will be explored in our next update, with the goal of making ethical reasoning capabilities more accessible across a wider range of AI systems.